(*iv*) A.C. Circuit Containing Inductance and Resistance in Series

Let an alternating emf

$$E = E_0 e^{j\omega t}$$

be applied to a circuit having an inductance L and a non-inductive resistance R in series (Fig. 13.10).

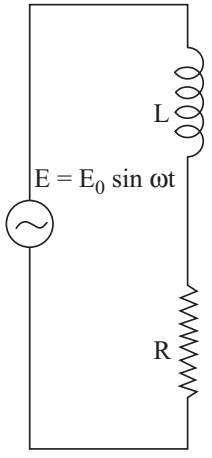


FIG. 13.10

The potential drop across the inductance is

$$V_L = j\omega LI$$

The potential drop across resistance is

$$V_R = RI$$

Here, I is the current at any instant t.

$\therefore$ $$E = j\omega LI + RI$$

Current in the circuit,

$$I = \frac{E}{R + j\omega L} \qquad ...(1)$$

But $$I = \frac{E}{Z} \qquad ...(2)$$

Impedance of R–L circuit,

$$Z = R + j\omega L \qquad ...(3)$$

$\therefore$ $$I = \frac{E_0 e^{j\omega t}}{\sqrt{(R^2 + \omega^2 L^2)} e^{j\theta}} \qquad \left(\text{where } \tan\theta = \frac{\omega L}{R}\right)$$

$$= \frac{E_0}{\sqrt{(R^2 + \omega^2 L^2)}} e^{j(\omega t - \theta)} \qquad ...(4)$$

$$= I_0 e^{j(\omega t - \theta)} \qquad ...(5)$$

Here, $$I_0 = \frac{E_0}{\sqrt{R^2 + \omega^2 L^2}} \qquad ...(6)$$

It represents the peak value of the current through the circuit.

The impedance Z of the circuit is given by the term

$$\sqrt{R^2 + \omega^2 L^2}.$$

The current lags in phase behind the emf by an angle

$$\theta = \tan^{-1}\frac{\omega L}{R}.$$

The variation of instantaneous values of emf and current with time are represented graphically in Fig. 13.11.

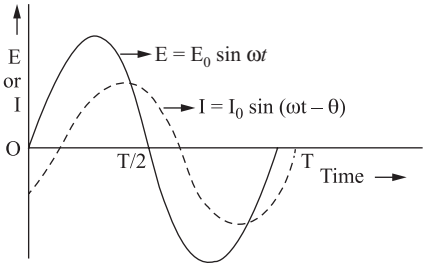


FIG. 13.11

(*v*) A.C. Circuit Containing Capacitance and Resistance in Series

Let an alternating voltage $E = E_0 e^{j\omega t}$ be applied to a circuit containing a resistance R and a capacitor C in series (Fig. 13.12). Let I be the current through the circuit and q, the charge on the plates of the capacitor at any instant. Then,

$$I\left(R + \frac{1}{j\omega C}\right) = E = E_0 e^{j\omega t} \qquad ...(1)$$

$$I = \frac{E}{R + \frac{1}{j\omega C}} \qquad ...(2)$$

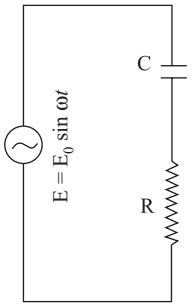


FIG. 13.12

But $$I = \frac{E}{Z} \qquad ...(3)$$

∴ Impedance of series *R*–*C* circuit,

$$Z = R + \frac{1}{j\omega C} = R - \frac{j}{\omega C}$$

$$I = \frac{E}{R - \frac{j}{\omega C}} = \frac{E_0 e^{j\omega t}}{\sqrt{\left[\left(R^2 + \frac{1}{\omega^2 C^2}\right)\right]} e^{-j\theta}} \qquad ...(4)$$

Here, $$\theta = \tan^{-1}\left(\frac{1/\omega C}{R}\right) \qquad ...(5)$$

∴ $$I = \frac{E_0}{\sqrt{\{R^2 + (1/\omega C)^2\}}} e^{j(\omega t + \theta)} \qquad ...(6)$$

The current in this circuit thus leads the applied voltage by an angle θ (Fig. 13.13).

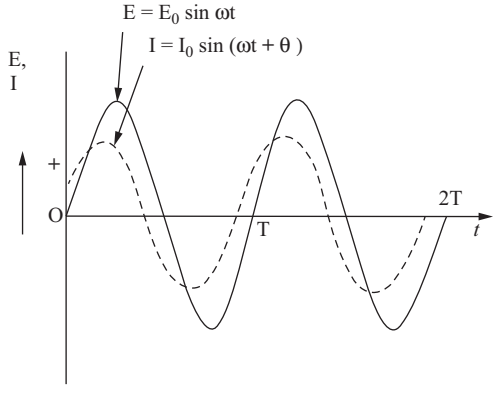


FIG. 13.13

The impedance of series *R*–*C* circuit, $Z = \sqrt{[R^2 + (1/\omega C)^2]}$.

The actual current in the circuit is $I = I_0 \sin(\omega t + \theta)$

(*vi*) Parallel Resonant Circuit

A parallel resonant circuit consisting of an inductance *L* and a capacitance *C* connected in parallel to the alternating e.m.f. is shown in Fig. 13.14.

Let the applied emf be

$$E = E_0 \sin \omega t$$

Current through the inductance lags behind the applied emf by $\frac{\pi}{2}$ and is given by,

P.D. across $R = I_{rms} \times R = 1.0 \times 100 =$ **100 volt (rms)**,

P.D. across $L = I_{rms} \times \omega L = 1.0 \times (100\pi) \times 5 =$ **1570 volt (rms)**

P.D. across $C = I_{rms} \times \dfrac{1}{C\omega} = 1.0 \times \dfrac{1}{(2 \times 10^{-6}) \times 100\pi}$

= **1570 volt (rms)**.

The voltages across the inductor and capacitor are much greater than the applied voltage. But they differ in phase by 180°. So their algebraic sum is zero.

13.4 PARALLEL RESONANT CIRCUIT

Here, capacitor C is connected in parallel to the series combination of resistance R and inductance L. The combination is connected across the AC source (Fig. 13.22). The applied voltage is sinusoidal, represented by

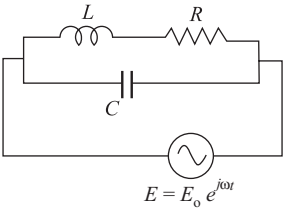


FIG. 13.22

$$E = E_0 e^{j\omega t}$$

Complex impedance of L - branch

$$Z_1 = \mathrm{R} + jL\omega$$

Complex impedance of C- branch

$$Z_2 = \frac{1}{jC\omega}$$

Z_1 and Z_2 are in parallel.

$$\frac{1}{Z} = \frac{1}{R + j\omega L} + \frac{1}{1/j\omega C} = \frac{1}{R + j\omega L} + j\omega C$$

$$= \frac{R - j\omega L}{(R + j\omega L) \times (R - j\omega L)} + j\omega C$$

$$= \frac{R}{R^2 + (L\omega)^2} + j\left[C\omega - \frac{L\omega}{R^2 + (L\omega)^2}\right]$$

The current $I = E/Z = E \times \dfrac{1}{Z}$

$\therefore$ $$I = E\left[\frac{R}{R^2 + (L\omega)^2} + j\left(C\omega - \frac{L\omega}{R^2 + (L\omega)^2}\right)\right]$$

Let $A \cos\phi = \dfrac{R}{R^2 + (L\omega)^2}$; $A \sin\phi = C\omega - \dfrac{L\omega}{R^2 + (L\omega)^2}$

$\therefore$ $$I = E\,(A\cos\phi + j\,A\sin\phi) = EAe^{j\phi} = E_0\,A\,e^{j(\omega t + \phi)}$$

where $$\phi = \tan^{-1}\frac{C\omega - \left(\dfrac{L\omega}{R^2 + (L\omega)^2}\right)}{\left(\dfrac{R}{R^2 + (L\omega)^2}\right)};$$

$$A^2 = \frac{R^2}{(R^2 + \omega^2 L^2)^2} + \left(C\omega - \frac{L\omega}{R^2 + \omega^2 L^2} \right)^2$$

The magnitude of the admittance

$$Y = \frac{1}{Z} = \frac{\sqrt{[R^2 + (\omega CR^2 + \omega^3 L^2\, C - \omega L)^2]}}{R^2 + \omega^2 L^2}$$

The admittance will be minimum, when

$$\omega CR^2 + \omega^3 L^2 C - \omega L = 0$$

or $$\omega = \omega_0 = \sqrt{\left[\frac{1}{LC} - \frac{R^2}{L^2} \right]}$$

or $$v_0 = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

This is the resonant frequency of the circuit.

If R is very small so that $\frac{R^2}{L^2}$ is negligible compared to $\frac{1}{LC}$,

$$v_0 = \frac{1}{2\pi \sqrt{(LC)}}$$

At such a minimum admittance, *i.e.*, maximum impedance, the circuit current is minimum.

The graph between current and frequency is shown in Fig. 13.23.

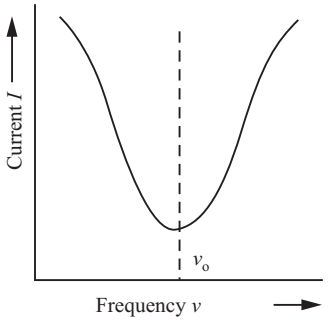


FIG. 13.23

Impedance at Resonance

At resonance, $$Z = \frac{R^2 + (L\omega)^2}{R}$$

But $$R^2 + (L\omega)^2 = \frac{L}{C} \text{ at resonance}$$

$\therefore$ $$Z = \frac{L}{RC}$$

Thus smaller the resistance R, larger is the impedance. If R is negligible, the mpedance is infinite at resonance.

Rejector Circuit. The parallel resonant circuit does not allow the current of the same frequency as the natural frequency of the circuit. Thus it can be used to suppress the current of this particular frequency out of currents of many other frequencies. Hence the circuit is known as a 'rejector' or 'filter' circuit.

Comparison between series and parallel resonant circuits

The behaviour of a parallel resonant circuit is strikingly different from that of a series resonant circuit. In both cases the impedance is resistive but, whereas parallel resonance implies maximum impedance, series resonance implies minimum impedance.